



UNIVERSITY OF
LINCOLN

MTH2006 Industrial and Financial Maths
PHY2003 Industrial and EconoPhysics

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2024-2025

Learning Outcomes

MTH2006

- ◆ Apply appropriate mathematics apparatus to a set of industrially relevant problems
- ◆ Solve appropriate mathematical problems following industrial problem-solving setting and timescales
- ◆ Present finding of relevant mathematical problems following industrial style and standards
- ◆ Apply basic tools of financial mathematics to describe selected phenomena of financial markets

Learning Outcomes

PHY2003

- ◆ Identify and apply physical laws and appropriate mathematics apparatus to a set of industrially relevant problems
- ◆ Solve appropriate physical problems following industrial problem-solving setting and timescales
- ◆ Present finding of relevant physical problems following industrial style and standards
- ◆ Apply basic tools of econophysics to selected phenomena of financial markets

Assignment Marks

	Assignments	Weight %	Final weight %	Components
Portfolio 1	Poster	70%	70%	I
	Poster Presentation	20%		
	Attendance	10%		
Assessment 1	Financial maths problems	100%	30%	II

Overview

- This is a challenging module where the time management skill is essential for completing this part of the module.
- Academic support is available through the module coordinator.
- Main aim: expose you to how maths or physics is used to solve “real world” problems outside universities.
- Working as an industrial mathematician / physicist involves jumping into new areas of mathematics / physics without fear, and teaching yourself something unfamiliar.
- **Read** the Module Handbook on Blackboard

Part 1: Industrial Mathematics /Physics

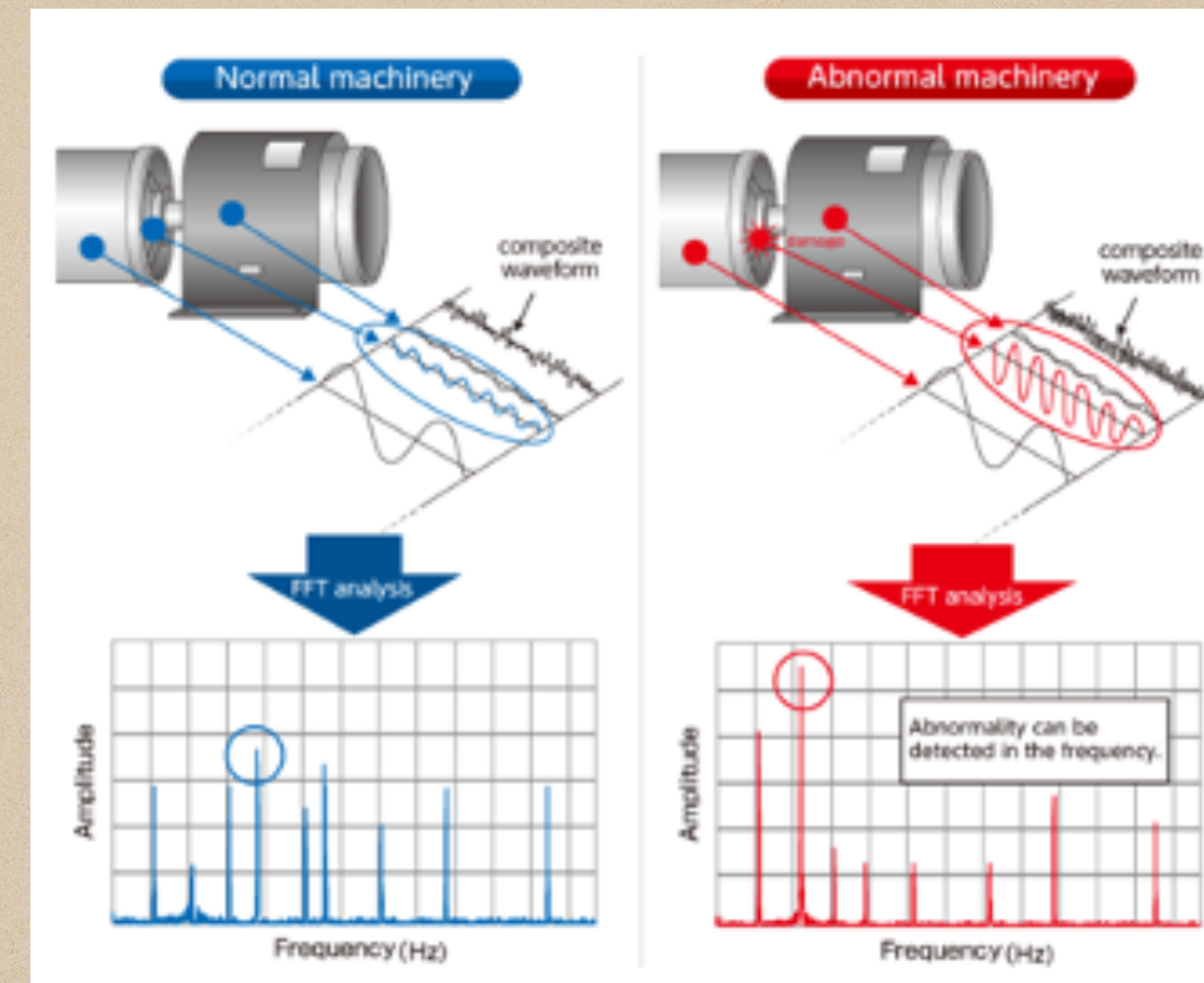
Some differences

- Mathematics students can choose to do your assessment on a mathematics or physics (applied maths).
- Physics students will need to still do a physics topic due to IOP accreditation and Learning Outcomes.

Part 1: Industrial Mathematics / Physics

- Lectures are given by staff and external speakers on industrial applications of mathematics / physics
 - Guest speakers from Industry are not paid
- Assessment:
 - You will **make a poster** about an industrial application of mathematics / physics.
You can choose any topics, but if you are in doubt, please ask the module coordinator.
 - Poster can be based on the topics of the talks.
 - You will **present** this poster in a simulated industry poster session.
 - **Attendance** will be recorded for the guest speakers, career sessions and workshops on sustainability.
An attendance of 100% will give a full mark for this component (7 sessions).
- Advice
 - Expect to be exposed to unfamiliar mathematics / physics.
 - Take detailed notes
 - Have fun

Part 1: Industrial Mathematics / Physics



- Fourier transforms are used to generate oceans in computer games and design / identify faulty parts in machines.

Part 1: Industrial Mathematics / Physics

Some examples of topics

- Algorithms for biometrics
- Mathematics for racing cars
- Algorithms for Machine Learning systems (applied to industry)
- Net - zero future (some mathematical/physical approach)
- Noise reduction
- Models for air traffic/car traffic management
- Mathematical models for ships
- Mathematics/physics in sports (example: rugby, tennis, football, etc)
- Interstellar travels
- Radiotherapy
- Fluid mechanics for sport improvements (example: swimming, racing cars etc.)
- Ideal orbits for satellites
- Maths in Agriculture
- Physics in tall buildings

Part 1: Industrial Mathematics / Physics

Careers Readiness Programme

Wednesday 5th February 11-12pm

Session 1: Discover your future; What are your options

Wednesday 19th February 11-12pm

Session 2: Helping you to build a professional network; it's not as scary as you might think!

Wednesday 5th March 11-12pm

Session 1: Get creative with you job search. Tapping into the hidden world of jobs online

Part 1: Industrial Mathematics / Physics

Sustanaibility workshops

Abstract

Your future employer will be thinking about sustainability! In these two sessions we will think first about the concept of sustainable development and how your skill set has prepared you to contribute to making the world a better place for people and the planet. We will use the United Nations Sustainable Development Goals as a framework for our discussion as they help us to focus on the breadth of the challenge of sustainability from health to energy. We will then go on to discuss how potential employers might be tackling sustainability in terms of what they say and what they are actually doing, and consider how you might contribute to that aspect of their mission.

Tuesday 4th February 11-12pm delivered by Prof Libby John

- **Session 1:** Sustainable Development needs you!

Wednesday 5th February 12-1pm delivered by Prof Libby John

- **Session 2:** What does your future employer contribute to sustainable development?

Part 1: Industrial Mathematics / Physics

Workshops

These workshops are designed for helping you to write/
finalize your poster (academic support).

Week 22: Tuesday 18th February 11-12pm

Week 24: Tuesday 4th March 11-12pm

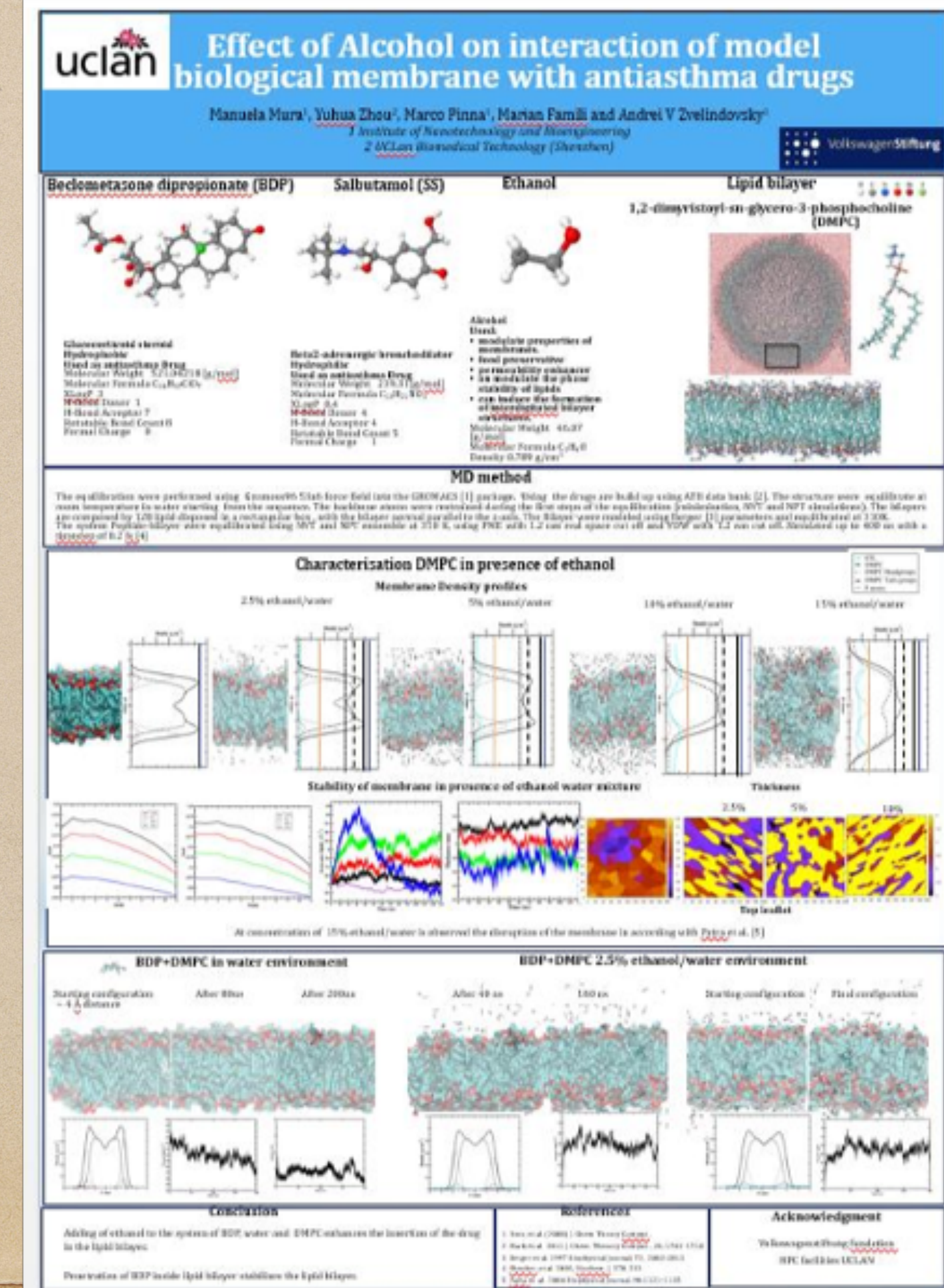
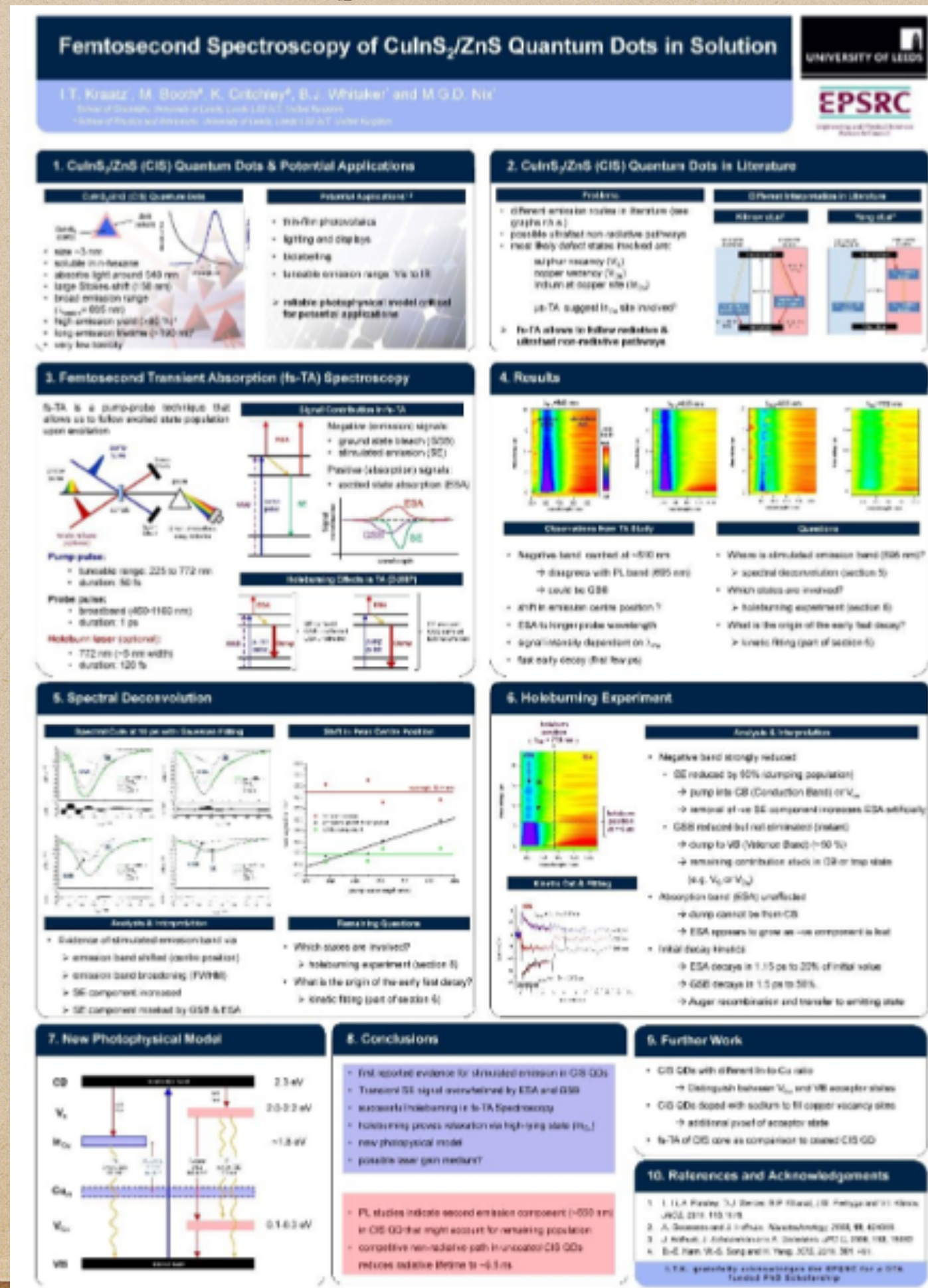
Part 1: Industrial Mathematics / Physics

Guest lectures

- ◆ Wednesday 29th of January 12-1pm — Jodie Renaud - NPL - **Quantum electrical metrology - who measures measurement**
- ◆ Wednesday 12th of February 11-1pm - Professor Natasha Maurits - Leiden Hospital
- ◆ Wednesday 19 th february 12-1pm - Hannah Dickens - Clinical Scientist - Lincoln Hospital-
A day in the life of a Clinical Scientist

Part 1: Industrial Mathematics / Physics

Example and templates of poster are given on Blackboard
Guidance is given in the Handbook



Part 1: Industrial Mathematics / Physics

- The poster should state the problem and how mathematics /physics is used solve it.
- The problem should relate to something in the real world with a clear example.

$$X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-i 2\pi k n/N}$$



How does a Fourier Transform help to efficiently render realistic waves in the game Sea of Thieves?

Part 1: Industrial Mathematics / Physics

Poster – Some simple rules to a good poster:

1. Define the purpose
2. Sell your work in ten seconds
3. The title is important
4. Many rules that apply to writing an abstract apply to designing a poster
5. Good posters have unique features
6. Layout and format are critical
7. Content should be concise
8. Posters should have a 'personality'
9. The impact of a poster happens during and after the presentation

Part 1: Industrial Mathematics / Physics

Poster – Some simple rules to a good poster:



Out of hundreds of posters at a conference why would people look at yours?

Part 1: Industrial Mathematics / Physics

Poster Format

Font must be easy to read at a distance

And don't make the text too small either

Use a variety of type sizes

to distinguish between different text elements of your poster

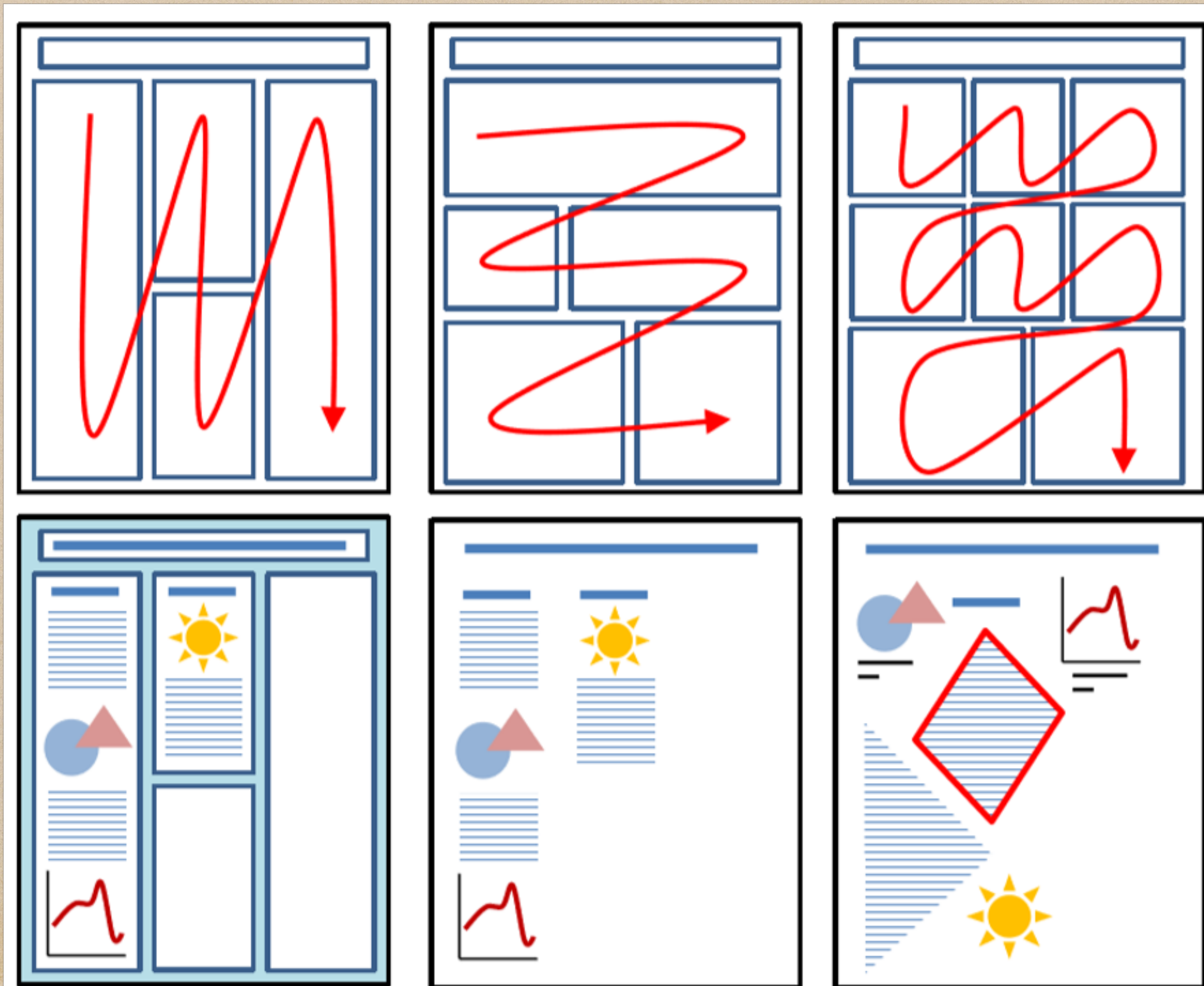
Using one and a half (at least) line spacing makes it much easier for the audience to read your text, especially if they have poor eyesight.

UPPER CASE TEXT CAN BE MORE
DIFFICULT TO READ FOR SOME
REASON

Ensure that the colour scheme helps the text to stand out

Part 1: Industrial Mathematics / Physics

Poster Layout



Part 1: Industrial Mathematics / Physics

	Bad Fail	Marginal Fail	III	Iiii	Ili	I	High First
A. Content of the poster							
The poster contains suitable material on industrial mathematics / physics, appropriate for 2nd year level and time available.							
The poster is well balanced: shows balance between breadth and depth of analysis, theory and examples.							
There is a logical structure to the poster.							
B. Clarity and appearance	-	-	-	-	-	-	-
Poster is attractive and professional. It is interesting with a good title. The poster is clear and easy to read.							
Presentation, including figures, style, grammar, spelling and use of IT. Correct use of references in text and acknowledgements in figure captions.							
The poster attracts the attention from several metres.							

Poster assessment criteria

Part 1: Industrial Mathematics / Physics

	Bad Fail	Marginal Fail	III	IIi	IIi	I	High First
A. Presentation Skills • Gains audience attention • Is relaxed & confident • Is easily heard							
B. Ability to summarise the poster • Gives a general introduction • Presentation is understandable to laymen • Poster content is put into context							
C. Ability to communicate • Concise delivery • Manner appropriate for broad audience • Effective handling of any questions							
D. Preparation and use of poster during presentation • Well-prepared • Refers to equations or diagrams on the poster							

Poster Presentation
assessment criteria

Part 1: Industrial Mathematics / Physics

Poster Presentation Assessment

- Poster presentations will take place 11th March in INB Atrium time 10am.
- You will need to print your poster in advance (A2) and submit electronic copy on Turnitin
- Your poster presentation should be 3 minutes in length.
- It should be professional, concise and compelling. In industrial settings, you often have very little time in which to pitch your ideas.
- You should be prepared to answer questions on your poster.
- Poster presentations differ from oral presentations in the fact that these are much less formal.
- The assessment is intended to evaluate your ability to summarise the poster and answer questions from the assessors.

Possible Key Skills

1. Quantitative and Analytical Skills:

- Solving complex problems using mathematical and physical principles.
- Analyzing data and creating models for real-world and industrial applications.

2. Critical Thinking and Problem Solving:

- Approaching challenges logically and creatively.
- Applying theoretical concepts to practical situations, including sustainability.

3. Research and Self-Learning:

- Exploring unfamiliar topics and synthesizing new knowledge independently.
- Adapting quickly to evolving tools, techniques, and concepts.

4. Time Management and Organizational Skills:

- Managing deadlines effectively for projects, assessments, and presentations.
- Prioritizing tasks and balancing academic commitments efficiently.

5. Communication and Presentation:

- Explaining complex ideas clearly through visual, written, and verbal formats.
- Tailoring content to diverse audiences, including professionals and laypeople.

6. Computational and Technical Skills:

- Using programming, simulations, and data visualization tools effectively.
- Bridging theory and practice with technology in innovative ways.

7. Understanding and Applying Sustainability:

- Linking mathematical and physical skills to the UN Sustainable Development Goals.
- Recognizing the ethical and societal impact of their work in sustainability-focused roles.

Dates

- Poster submission date: 6th of March (To be submitted via Turnitin in PDF format)
- Poster Presentation is in week 25 Session 11th March and is in INBatrium, time 10am.
The poster should be printed in advance to be ready for display on the 11th of March.
- Financial Mathematics lectures start in week 27

Above dates are to give you a rough idea of key dates, always check final dates on Blackboard.

Questions?